

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

Group ID:

Project Title:

1. IT22732736: Knowledge graph Module
2. IT22348784: Explainable Legal Reasoning Module
3. IT22113504: Visual integrated multi-perspective legal summarizer
4. IT22344892: Bias & Risk evaluator

		IT22732736	IT22348784	IT22113504	IT22344892
Data Pipeline	Data Sources:	154 Sri Lankan property deed PDFs (sale, gift, mortgage, partition, testamentary types) across 5 districts (Colombo, Kandy, Galle, Jaffna, Kurunegala), 1990–2024. Corpus: 89 authentic research specimens + 42 Colombo notarial samples + 23 synthetic deeds for underrepresented types.	21 Sri Lankan statutory instruments downloaded from SriLankaLaw.lk (official Govt portal) covering 8 property law topic areas. Additional: 35 legal term definitions, 5 standard deed templates (sale, gift, mortgage, partition, testamentary), 18 foundational legal principles — all compiled into a structured JSON knowledge base.	Property law deeds collected from practicing lawyers (Galle & Colombo), including deeds of sale, leases, gifts, wills, and transfer agreements, along with supplementary legal documents from Sri Lankan legal websites.	Property law deeds collected from practicing lawyers, Property law clauses derived from real-world legal documents, with a synthetic dataset of 580 clauses generated using Gemini 2.5 Pro to overcome data scarcity.
	Data Preprocessing:	pdfplumber (primary layout-aware extraction); PyPDF2 (fallback); pytesseract OCR (scanned docs with denoising, deskewing, contrast)	Statute text: HTML stripped via BeautifulSoup; whitespace collapsed; ligature/OCR artefacts corrected; statute segmented into numbered sections via regex. Definition entries	Text extraction from PDFs using tools like PyMuPDF/pdfplumber, followed by cleaning (noise removal,	Text extraction from PDF/DOCX/TXT files, clause segmentation using NLTK, custom abbreviation handling

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		enhancement). Text normalisation: ligature fix, whitespace collapse, line break removal, OCR artefact normalisation.	normalised with statute+section metadata. Each entry validated (all 8 JSONL fields present, ≥ 3 reasoning steps, ≥ 1 legal reference, IRAC keys complete). Entries failing validation regenerated with failure as negative prompt example.	whitespace normalization, punctuation standardization), clause segmentation using regex, and legal jargon normalization using a CSV-based dictionary.	to prevent sentence break errors, and regex-based metadata stripping to remove clause numbers and headings.
	Data storage and versioning	Structured JSON per deed stored in data/extracted/. SpaCy .spacy DocBin files for train/dev splits. Neo4j Aura cloud graph database (9 node types, 11 relationship types). Model artefact at deed_ner_model1/. Batch processing log per deed.	Statutes: data/sri_lankan_laws/processed/ (JSON with URL, timestamp, hash). Knowledge base: data/knowledge_base/ (definitions.json, deed_templates.json, legal_principles.json, knowledge_base.json). Dataset: finetuning_data.jsonl (2,243 IRAC examples, OpenAI messages format). Train/val/test split (80/10/10, seed=42). Model: HuggingFace Hub — Sivanuja/Legal_vision_llama-3.1-8B-instruct-v1. Dataset: Sivanuja/Legal_vision_finetuning_data.	Structured datasets stored as CSV/Excel files (clauses table, multi-perspective summaries, legal jargon dictionary), processed using Pandas DataFrames; traceability maintained via clause IDs and dataset linkage. GitHub is used for version control and management of code, datasets, and model updates.	Structured datasets stored in Excel/CSV format and processed as tensors for model training; GitHub used for version control of code, datasets, and model updates.
Model Development	Model Selection	SpaCy v3 transition-based NER (BIOES scheme, tok2vec CNN width=96, depth=4). Hybrid: NER + 43 regex patterns merged via confidence-weighted	Meta LLaMA 3.1 8B Instruct selected from 5 candidates (LLaMA 3.1 8B, Mistral 7B v0.3, SaulLM-7B, LLaMA 3.1 70B, Gemma 2 9B) based on: superior instruction-following, 128K context	LegalBERT (for clause classification), LegalLED (Longformer Encoder-Decoder) as the primary	Dual-model architecture: DistilBERT for bias classification

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		algorithm. Rejected: LEGAL-BERT (insufficient South Asian data), LexNLP (US-only).	window, open community licence, full Unsloth integration (2x throughput). QLoRA (Dettmers et al., 2023): NF4 4-bit quantisation of frozen base (~4GB) + LoRA rank-16 adapters on all 7 projection layers (q,k,v,o_proj; gate,up,down_proj) — <1% of 8B params trainable. SaulLM-7B retained as evaluation-only comparison (not fine-tuned) to isolate jurisdiction-specific fine-tuning advantage.	summarization model, with T5 used during experimentation for performance comparison. spaCy is used for NLP tasks such as NER and fallback extraction.	Legal-BERT for risk severity classification (selected due to better understanding of legal terminology)
	Model Training	50 iterations on 123-deed training split (~2.5 hrs GPU). Adam lr=0.001, polynomial decay. Dropout=0.1. Compounding batch 4→1000. 21 entity types. Final F1=99.53% (Precision 99.19%, Recall 99.87%). Loss: 37,838→181 (99.5% reduction).	GPT-4o teacher-student synthesis: 2,243 IRAC Chain-of-Thought examples (8 topic areas, diverse scenarios/parties/regions). QLoRA fine-tuning: 3 epochs, 325 steps, ~45 min on A100 40GB. Config: lr=2e-4 cosine, batch=2, grad accum=4 (eff. batch=8), max_seq=2048, warmup=0.03, AdamW 8-bit paged, LoRA dropout=0.05, LoRA $\alpha=16$. Framework: HuggingFace TRL SFTTrainer + Unsloth. Training loss: 1.82→0.29; val loss: 0.53→0.37 (monotonic, no overfitting). Results: BLEU 52.63, ROUGE-1 0.707, ROUGE-2 0.472, BERTScore F1 0.931 — outperforms base LLaMA (+223% BLEU) and base SaulLM (+583% BLEU).	LegalBERT was fine-tuned on clause-level labeled data (4 perspectives). For summarization, multiple models (T5 and LegalLED) were evaluated, and LegalLED was selected due to better performance on long legal documents, maintaining context and reducing information loss. Training and evaluation were conducted using ROUGE metrics.	Fine-tuning DistilBERT (5-class bias classification) and Legal-BERT (3-class risk severity) with extended token length (256 tokens), optimized learning rates, and multiple training epochs to improve accuracy.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

Model Integration	Tools used	SpaCy v3 (NER), pdfplumber/PyPDF2/pytesseract (PDF extraction), Neo4j 5.x Aura + Python Driver (graph DB), OpenAI GPT-4o (GraphRAG NL response), FastAPI 0.110.x + Uvicorn (REST backend), React 18.x (frontend), LLaMA 3.1 8B fine-tuned (Reasoning Module via RAG context object).	Data collection: Python requests + BeautifulSoup (statute scraping), sri_lankan_law_downloader_v2.py. Dataset generation: OpenAI GPT-4o API (teacher), legal_reasoning_dataset_generator.py (JSON validation + JSONL conversion). Fine-tuning: HuggingFace TRL SFTTrainer, Unsloth (2x speedup via custom CUDA kernels), PEFT (LoRA), bitsandbytes (NF4 quantisation), Google Colab A100. Evaluation: SacreBLEU 2.3.1, rouge-score 0.1.2, bert-score 0.3.13 (bert-base-uncased backbone). Deployment: FastAPI + Render.com (API), React 18 + Vercel (frontend), HuggingFace Hub (model + dataset). RAG integration: REST context object interface with Sharan's Neo4j GraphRAG (statutes, transactions, parties pathways).	Python, HuggingFace Transformers, spaCy, PyMuPDF, pdfplumber, Pandas, regex (re), Matplotlib, Plotly, Chart.js, FastAPI/Flask, React, and Gemini API for infographic generation.	Python, HuggingFace Transformers, PyTorch, NLTK, Regex (re), Flask, PyPDF2, python-docx, and Gemini API for dataset generation.
Model Deployment	Testing Environments	NER: 31-deed held-out test set (SpaCy evaluate, F1>95% threshold). Extraction: 30 held-out deeds, mandatory field coverage >95%. Graph: Cypher count queries + MERGE injection test + 5 REFERS_TO_PRIOR chain	Data pipeline: 21 statute URLs (HTTP 200 + char count >1,000 each). KB schema validation + 10% human review. Dataset: 99.8% schema pass (2,239/2,243); 97.1% legal accuracy on 20% human sample (436/449); 99.1% structural completeness. Tokenisation: all 2,243 examples ≤2,048 tokens. Training: loss monotonic (325 steps, no	Functional testing (clause segmentation, classification, summarization), performance testing (processing time), usability testing with non-expert users, and evaluation using	Functional testing (clause segmentation and classification), performance testing, and evaluation using Accuracy, Precision, Recall, and F1-score metrics.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		tests. GraphRAG: 30-query intent test (93.3% accuracy) + 10 GPT-4o response cycles. Upload: 10 unseen deeds via React UI (<30s criterion). Frontend: 11 modules + mobile 375px.	NaN/divergence); val loss never exceeds train loss; LoRA params verified (79,953,664 / 8,030,261,248 = 0.996%). Evaluation fairness: identical 225-example test split, identical generation params, pinned library versions. Deployment: FastAPI health check HTTP 200, <15s latency; mobile responsive (iPhone Safari, Android Chrome, 375px viewport).	metrics such as Accuracy, F1-score, and ROUGE.	
	Deployment Platform	Frontend: Vercel (legal-insights-hub.vercel.app). Backend API: Render.com (FastAPI). Graph DB: Neo4j Aura cloud (bolt protocol). Reasoning Module: Render.com (legal-vision-api.onrender.com).	Model: HuggingFace Hub (Sivanuja/Legal_vision_llama-3.1-8B-instruct-v1). Training Dataset: HuggingFace Datasets (Sivanuja/Legal_vision_finetuning_data). Inference API: Render.com (legal-vision-api.onrender.com — FastAPI, NF4 base + LoRA adapter loaded at startup). Frontend: Vercel (legal-insights-hub.vercel.app — shared Legal Insights Hub with Sharan's KG component)	Web-based platform using FastAPI/Flask backend with React frontend, running in a Python-based environment.	Flask-based web application with REST API and interactive frontend interface.
	Deployment Method	Batch load: neo4j_loader_v2.py processes all extracted JSONs in transactional batches (50 deeds/commit); 154 deeds load in <2 min. Real-time: POST /api/v1/upload-deed → HybridExtractor → Quality	Three REST endpoints: POST /api/v1/reason (accepts question + optional KG context object {statutes, transactions, parties}; constructs system/user prompt; inference temp=0.1, max_new_tokens=1024, do_sample=False; returns IRAC response). POST /api/v1/analyze-deed (deed text → legal validity analysis). GET	End-to-end pipeline deployment with document upload → processing → summarization → visualization, integrated via APIs and interactive dashboards.	End-to-end pipeline deployment with file upload → preprocessing → dual-model inference → Trust Score calculation → interactive visualization dashboard.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report

		Scorer → Neo4j conditional load (EXCELLENT/GOOD auto-load; REVIEW/POOR queued). New deeds queryable within 15 seconds.	/api/v1/decision-tree/{topic} (structured decision tree JSON for frontend flowcharts). Model loaded once at startup (NF4 base + LoRA adapter merge). RAG integration: KG component calls GraphRAG, assembles context object, forwards to /reason endpoint — each component independently deployable.		
Future Enhancements	Model Improvement:	(1) Expand corpus to 500+ deeds (all 9 provinces, 1840–present, Tamil-language elements) → target PARTY_VENDEE F1 ≥95%. (2) Replace keyword intent detector with trained classifier (target 98%+ accuracy). (3) Add StatutoryProvision + case law graph nodes. (4) Gazette notification parsing for near-real-time updates.	(1) Dataset expansion: 2,243 → 5,000+ examples incorporating Sri Lanka Law Reports, Supreme Court & Court of Appeal property decisions, Land Registration Tribunal decisions — especially for multi-statute reasoning. (2) Human evaluation by 3–5 qualified Sri Lankan property law advocates (planned within 6 months of submission). (3) Full KG RAG integration deployment (property-specific ownership chain grounding via Sharan's Neo4j). (4) Multilingual extension: Sinhala and Tamil IRAC examples via GPT-4o teacher synthesis. (5) Active learning flywheel: user query annotation pipeline for continuous dataset expansion. (6) Robustness testing: adversarial queries, out-of-distribution topics, post-training statutory amendments.	Expand to other legal domains (family law, criminal law, contracts), improve generative summarization flexibility, enhance visualization accuracy, and support multilingual and international scalability.	Improve dataset size and real-world coverage, enhance classification accuracy, extend to multiple jurisdictions, and refine Trust Score algorithm for better explainability and robustness.

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

MLOPs Report
